

Practice

Tuesday, 8 September 2026 10:01 am

Model the "reorder past meal" process as a BPMN flowchart.

Write 5 SQL SELECT queries against a sample database (use a free sample DB like Northwind) to answer business questions like "top 5 customers by order value."

Reorder
flowchart

Reorder_Pas
t_Meal_B...

1. Top 5 customers by order value

```
SELECT TOP (5)
    c.CompanyName,
    SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)) AS TotalSales
FROM Customers c
JOIN Orders o ON c.CustomerID = o.CustomerID
JOIN [Order Details] od ON o.OrderID = od.OrderID
GROUP BY c.CompanyName
ORDER BY TotalSales DESC;
```

Customers(+) 1 X			
SELECT TOP (5) c.CompanyName, SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)) AS TotalSales			
Grid	AZ CompanyName	123 TotalSales	
1	QUICK-Stop	110,277.304977417	
2	Ernst Handel	104,874.9787139893	
3	Save-a-lot Markets	104,361.9499206543	
4	Rattlesnake Canyon Grocery	51,097.8003330231	
5	Hungry Owl All-Night Grocers	49,979.9050006866	

2. Top 5 best-selling products

```
SELECT TOP (5)
    p.ProductName,
    SUM(od.Quantity) AS TotalUnitsSold
FROM Products p
JOIN [Order Details] od ON p.ProductID = od.ProductID
GROUP BY p.ProductName
ORDER BY TotalUnitsSold DESC;
```

Products(+) 1 X			
SELECT TOP (5) p.ProductName, SUM(od.Quantity) AS TotalUnitsSold			
Grid	AZ ProductName	123 TotalUnitsSold	
1	Camembert Pierrot	1,577	
2	Raclette Courdavault	1,496	
3	Gorgonzola Telino	1,397	
4	Gnocchi di nonna Alice	1,263	
5	Pavlova	1,158	

3. Sales by employee

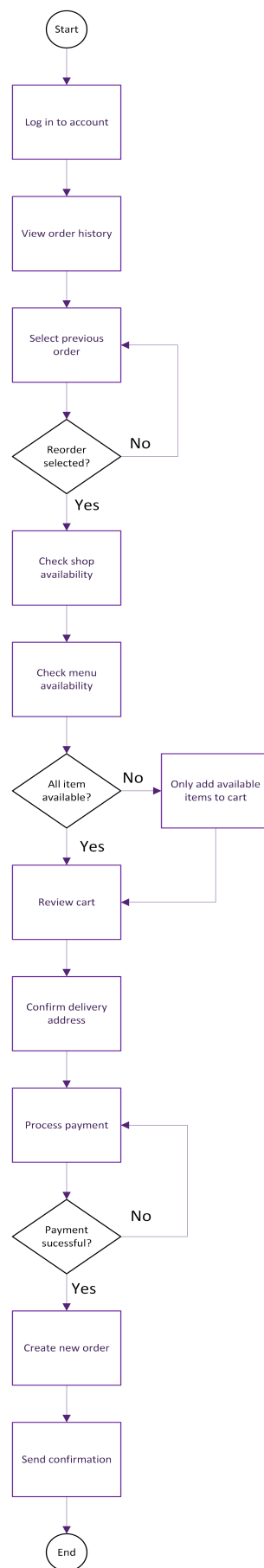
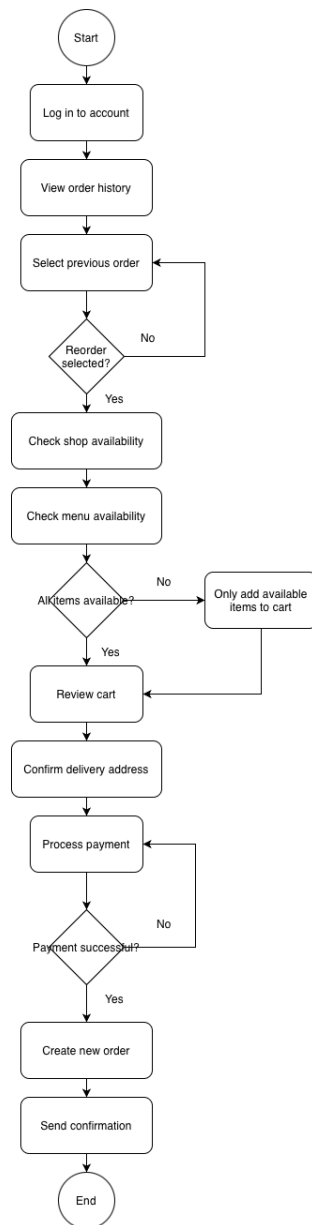
```
SELECT
    e.FirstName + ' ' + e.LastName AS EmployeeName,
    SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)) AS TotalSales
FROM Employees e
JOIN Orders o ON e.EmployeeID = o.EmployeeID
JOIN [Order Details] od ON o.OrderID = od.OrderID
GROUP BY e.FirstName, e.LastName
ORDER BY TotalSales DESC;
```

Employees(+) 1 X			
SELECT e.FirstName + ' ' + e.LastName AS EmployeeName, SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)) AS TotalSales			
Grid	AZ EmployeeName	123 TotalSales	
1	Margaret Peacock	232,890.8459472656	
2	Janet Leverling	202,812.8427934647	
3	Nancy Davolio	192,107.6043205261	
4	Andrew Fuller	166,537.7549781799	
5	Laura Callahan	126,862.2777042389	
6	Robert King	124,568.2353820801	
7	Anne Dodsworth	77,308.0667095184	
8	Michael Suyama	73,913.1292438507	
9	Steven Buchanan	68,792.2825393677	

4. Which customers place the most orders

```
SELECT TOP (10)
    c.CompanyName,
    COUNT(o.OrderID) AS NumberOfOrders
FROM Customers c
JOIN Orders o ON c.CustomerID = o.CustomerID
GROUP BY c.CompanyName
ORDER BY NumberOfOrders DESC;
```

Customers(+) 1 X			
SELECT TOP (10) c.CompanyName, COUNT(o.OrderID) AS NumberOfOrders			
Grid	AZ CompanyName	123 NumberOfOrders	



	Companyname	NumberOfOrders
1	Save-a-lot Markets	31
2	Ernst Handel	30
3	QUICK-Stop	28
4	Folk och fä HB	19
5	Hungry Owl All-Night Grocers	19
6	Berglunds snabbköp	18
7	HILARION-Abastos	18
8	Rattlesnake Canyon Grocery	18
9	Bon app'	17
10	Frankenversand	15

5. Revenue by product category

```
SELECT
  c.CategoryName,
  SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)) AS TotalRevenue
FROM Categories c
JOIN Products p ON c.CategoryID = p.CategoryID
JOIN [Order Details] od ON p.ProductID = od.ProductID
GROUP BY c.CategoryName
ORDER BY TotalRevenue DESC;
```

	CategoryName	TotalRevenue
1	Beverages	267,868.1805229187
2	Dairy Products	234,507.2852172852
3	Confections	167,357.2248315811
4	Meat/Poultry	163,022.3590888977
5	Seafood	131,261.7374248505
6	Condiments	106,047.0849895477
7	Produce	99,984.5800685883
8	Grains/Cereals	95,744.587474823

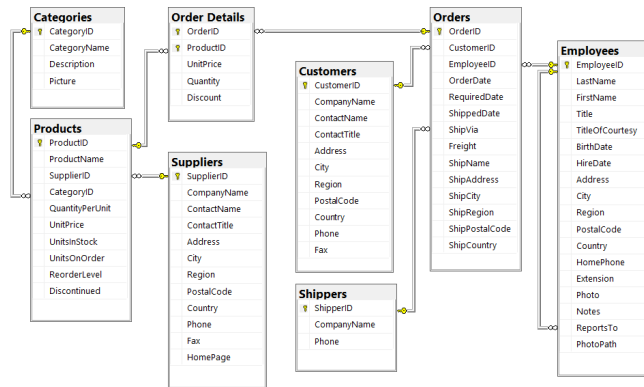


Figure 10.1: The Northwind database tables and relationships

<https://github.com/microsoft/sql-server-samples/tree/master/samples/databases/northwind-pubs>